Healthcare Analytics 5 (2024) 100341

Contents lists available at ScienceDirect

Healthcare Analytics

journal homepage: www.elsevier.com/locate/health

A deterministic compartment model for analyzing tuberculosis dynamics considering vaccination and reinfection

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ARTICLE INFO

Keywords:
Tuberculosis control analytics
Reinfection
Vaccination
Backward bifurcation
Sensitivity analysis

ABSTRACT

Tuberculosis is a pressing global health concern, particularly pervasive in many developing nations. This study investigates the influence of treatment failure on tuberculosis control strategies, incorporating vaccination interventions using a deterministic compartmental epidemiological model. Mathematical analysis unveils disease-free and endemic equilibrium points, with the control reproduction number determined using next-generation methods. Identifying endemic equilibrium points and determining the control reproduction number provide essential metrics for assessing the effectiveness of control strategies and guiding policy decisions. The model exhibits a backward bifurcation phenomenon, leading to multiple endemic equilibria despite a reproduction number below one due to reinfection. Sensitivity analysis using Latin Hypercube Sampling/Partial Rank Correlation Coefficient elucidates parameter impacts on the control reproduction number. Vaccination efficacy is crucial for quality and validity, with superior quality and longer validity yielding more significant effects. While reinfection may not directly affect the reproduction number, its influence is pivotal in determining tuberculosis persistence or extinction. This study underscores the intricate interplay of factors in tuberculosis control strategies, providing insights vital for effective interventions and policy formulation.

1. Introduction

Tuberculosis (TB) is a severe global health problem caused by the bacteria Mycobacterium tuberculosis [1,2]. Primarily, these bacteria target the respiratory system, particularly the lungs [3]. However, tuberculosis can also affect other organs such as the kidneys, spine, brain, and nerves [4,5]. Transmission of tuberculosis occurs through the inhalation of droplets containing the bacteria, and individuals infected with tuberculosis often experience persistent cough, fever, chest pain, and other symptoms. If left untreated, tuberculosis can be fatal [6]. According to the latest data from the World Health Organization (WHO), it is projected that there be approximately 10 million new cases of tuberculosis worldwide in 2021 [7], resulting in over 1.5 million deaths [8]. Despite the availability of a cure, tuberculosis remains a leading cause of death, following closely after COVID and HIV/AIDS [9].

Preventing tuberculosis infection and halting the progression from infection to disease are crucial in reducing the incidence of tuberculosis to the desired level [9]. The primary healthcare intervention for achieving this reduction is tuberculosis preventive treatment, which typically lasts for nine months [7,10]. Other methods of tuberculosis prevention and control include vaccination with the Bacille Calmette–Guérin (BCG) vaccine, early detection and timely treatment of patients, and screening for high-risk groups [11]. While it has long been known that BCG vaccination does not provide protection against tuberculosis infection, it does offer some defense against the development of active disease, BCG vaccination has proven to be effective and efficient in treating children with tuberculosis. However, its efficacy in adults is limited as it only provides partial protection against respiratory. Therefore, the development of a new vaccine targeting both children and adults is necessary [12,13].

Several deterministic mathematical models have been developed to study the dynamics of tuberculosis transmission in populations, considering factors such as vaccination. For instance, Sulayman et al. [14] extended a deterministic mathematical model for tuberculosis transmission dynamics to examine the impact of imperfect vaccines and other factors, including reinfection among treated individuals and exogenous reinfection. They discovered that both the number of susceptible individuals vaccinated and the efficacy of the vaccine are equally significant in reducing the burden of tuberculosis. In a recent study by Li et al. [11], a tuberculosis model was constructed, incorporating factors such as varying development rates, vaccination targeting susceptible individuals to reduce the likelihood of infection, incomplete

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https://doi.org/10.1016/j.health.2024.100341
Received 26 August 2023; Received in revised form 26 April 2024; Accepted 1 May 2024
Available online 7 May 2024
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treatment, and relapse. Their analysis revealed that the tuberculosis-free and endemic equilibriums were globally stable when the basic reproduction number was less than and greater than one, respectively.

Mathematical transmission models are widely used to understand how disease may spread in a population [15–18], including in tuberculosis. Tuberculosis is one of a most complex diseases, characterized by the complexities arising from reinfection and relapse dynamics. Moreover, the emergence of multidrug-resistant TB (MDR TB) further exacerbates the challenges in TB management and control efforts. Central to understanding TB's intricate dynamics is the recognition of its slow–fast progression, a hallmark feature that poses unique challenges for treatment and control strategies. Many mathematical models have been introduced by authors to understand the mentioned phenomena. Authors in [19–23] develop their model considering reinfection. They found that reinfection contributing to the ongoing circulation of the bacterium within populations and make the control effort become more unpredictable. Authors in [11,24] introduced their tuberculosis model incorporating slow–fast progression. In response to these complexities, interventions such as vaccination and treatment have gained prominence as the cornerstone strategies for combating TB on a global scale, reflecting the growing recognition of the need for comprehensive approaches to address the multifaceted challenges posed by the disease. Numerous studies have been conducted on tuberculosis transmission models employing various strategies, such as vaccination, treatment, and early diagnosis [3,8,25,26] and most recent literature in [27–29]. Authors in [27] introduced a time-delay treatment in their tuberculosis model incorporating age-dependent latency. The study presents an enhanced strategy aimed at achieving the World Health Organization's global goals for TB control. In [28], a tuberculosis mathematical model incorporating age structure, vaccination, and treatment dynamics was developed and parameterized using incidence data from China. Optimal control strategies were then applied to identify the most effective interventions for TB control. Asymptotic cases included by authors in [29] with an intervention of vaccination as a control effort. Sensitivity analysis were given to show the effect of vaccination in lowering tuberculosis transmission. In our recent work in [30], we develop a mathematical model for tuberculosis involving slow–fast progression and limited hospital bed capacity. Our analytical and numerical results shows a possibility when tuberculosis may still persist even the reproduction number is small than one. In some cases, this phenomena triggered by a backward bifurcation appearance at basic reproduction number equal to one. Please see [31–33] for another example of backward bifurcation appearance in other epidemiological models.

Emphasizing the gravity of Tuberculosis, it is essential for many researchers to discuss the links between tuberculosis and other infectious diseases such as SARS-CoV-2 or HIV, and their potential impacts on public health. Individuals co-infected with tuberculosis and either SARS-CoV-2 or HIV face increased health risks and complications due to compromised immune systems. The presence of tuberculosis can exacerbate the spread and severity of COVID-19 or HIV, and vice versa. Disruptions in tuberculosis diagnosis and treatment services during the COVID-19 pandemic have further heightened concerns about tuberculosis transmission and incidence rates. Therefore, understanding the intersections between tuberculosis and other infectious diseases is crucial for implementing comprehensive public health measures to mitigate their spread and impact. There are several articles discussing about the co-infection between tuberculosis and other diseases, such as HIV [34,35], malaria [36], and recently with COVID-19 [20,37]. On these research, authors try to find an insight from mathematical approach about the interaction between tuberculosis and other diseases in a population scale. They mainly use a concept of the basic reproduction number, control reproduction number, or sometimes with invasion reproduction number.

While many articles discuss the dynamics of tuberculosis using compartmental models, only a few of them incorporate temporary immunity. To address this gap, we developed a tuberculosis transmission model that considers the effects of vaccination and reinfection by incorporating temporary immunity. This model enables a better understanding of the interactions among infected individuals, vaccinated individuals, and those who experience reinfection. By utilizing this model, we can identify factors that influence tuberculosis transmission, evaluate the effectiveness of vaccination, and assess the impact of reinfection on the spread of the disease.

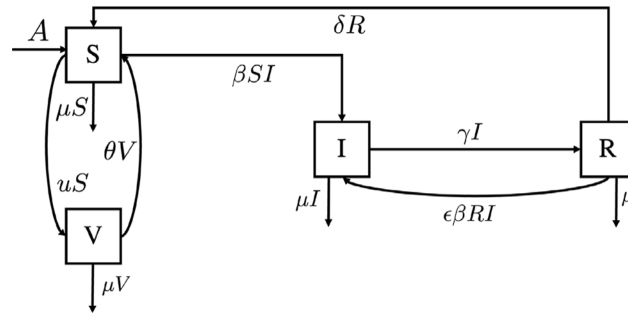

Fig. 1. Transmission diagram of tuberculosis model in system (1).

This article divided into five sections. In Section 2, we describe the construction of the tuberculosis transmission model with vaccination. Positiveness criteria, Non-dimensionalization of variables and parameters, Equilibrium and reproduction number, and local stability criteria are presented in Sections Section 3. Additionally, Section 4 includes numerical experiments, and finally, in Section 5, we provide our concluding remarks.

2. Model construction

This article presents a deterministic compartment model for studying tuberculosis dynamics that incorporates vaccination and reinfection. This model divides the population into four compartments, namely the susceptible (S), vaccinated (V), infected (I) and recovered (R). The total population is denoted as N, where $N = S + V + I + R$. There are several important assumptions used to construct the model as follows:

- **(A1)**: All newborn (A) are susceptible. Considering that infants born to untreated TB-infected women may have lower birth weights and, in rare cases, be born with tuberculosis, it can be inferred that all newborns are potentially susceptible to tuberculosis. Moreover, despite the drugs used in the initial tuberculosis treatment regimen crossing the placenta, there is no evidence indicating harmful effects on the fetus.
- **(A2)**: Tuberculosis vaccine is not for lifetime use. From many reports [38,39], it is estimated that current widely used vaccine for TB, i.e Bacille Calmette–Guerin (BCG), can give a protection between 10-15 years before it gradually decrease it effectiveness. In our model, we use u to represent the rate of vaccination and θ represent the rate of vaccine wane.
- **(A3)**: Tuberculosis vaccine assumed can be used for infants, young children and adults. The Bacille Calmette–Guérin (BCG) vaccine is primarily administered to infants and young children in countries where TB is prevalent. While BCG vaccination can provide some protection against severe forms of TB in children, its efficacy in preventing pulmonary TB in adults is variable and limited.
- **(A4)**: Reinfection of tuberculosis. Reinfection refers to acquiring a new infection after being exposed to either the same or potentially different strains of TB bacteria [40,41]. We assume that the probability of successful of infection for reinfection is smaller than primary infection.

Table 1
Description of parameters in system (1).

Parameter	Description	Range	Ref.
A	Recruitment rate of human	$\frac{1000}{65\times 12}$ $\frac{\text{human}}{\text{month}}$	Estimated
β	Probability of successful infection rate	$[0.01,\ 5]$ $\frac{1}{\text{year}\times\text{human}}$	[42–45]
ϵ	Correction parameter for reinfection term	$[0, 1]$	Assumed
u	Rate of vaccination	$[0,1]$ $\frac{1}{\text{day}}$	[46]
θ	Vaccine wane rate	$[0.067,\ 0.1]$ $\frac{1}{\text{day}}$	[47–50]
γ	Recovery rate	$[0.5,2]$ $\frac{1}{\text{year}}$	[30,51]
δ	Drop out rate due to waning immunity	$[0.003,\ 0.02]$ $\frac{1}{\text{day}}$	[30,52]
μ	Natural death rate	$\frac{1}{65\times 12}$ $\frac{1}{\text{month}}$	Estimated

Using previous assumptions and the model presentation, illustrated in Fig. 1, The model construction is given as follows. We start by assuming that all newborns are born susceptible at a constant rate A. Additionally, we consider a constant vaccination rate u, representing the movement of individuals from the susceptible class to the vaccinated compartment. Given that tuberculosis vaccines offer finite protection, there exists a constant dropout rate (θ) from the vaccinated compartment back to the susceptible compartment. Assuming full protection against tuberculosis infection, only susceptible individuals can contract the disease through direct contact with those in the infected compartment, occurring at a constant infection rate β. Individuals in the infected compartment recover at a constant rate γ. Considering the possibility of reinfection, individuals in the recovered compartment may be re-infected by those in the infected compartment, with a rate of $\epsilon\beta$, where ϵ serves as a correction parameter ranging between zero and one. Furthermore, individuals may transition back to the susceptible compartment from the recovered compartment due to waning immunity, occurring at a constant rate δ. Lastly, each compartment experiences a decrease due to the natural death rate μ.

Based on above model description, the mathematical model for tuberculosis under the impact of vaccination and reinfection is given by the following system of ordinary differential equations:

$$\frac{dS}{dt} = A - \beta SI + \delta R - \mu S - uS + \theta V, \tag{1a}$$
$$\frac{dV}{dt} = uS - \theta V - \mu V, \tag{1b}$$
$$\frac{dI}{dt} = \beta SI + \epsilon\beta RI - \gamma I - \mu I, \tag{1c}$$
$$\frac{dR}{dt} = \gamma I - \epsilon\beta RI - \delta R - \mu R, \tag{1d}$$

completed with a non-negative initial conditions. Please see Table 1 for the parameter description, it value range, and references for it.

3. Model analysis

In this section, we will explore the behavior of the model. We will start by analyzing the Positiveness criteria, existence of equilibria and calculating the basic reproduction number. Finally, we will examine the local stability criteria of the model equilibrium.

3.1. Positiveness criteria

Before delving into the subsequent section regarding the existence of equilibrium and the basic reproduction number form, it is important to establish an initial result concerning the positive and invariant feasible region. The model presented above describes the dynamics of the human population, and it is expected that all variables in the model, being of an epidemiological nature, must be non-negative. Consequently, it is predictable that there exists a feasible region within which the model holds epidemiological interpretation.

Theorem 1. *System (1) is positively invariant in the following region.*

$$\Omega = \{(S, V, I, R) \in \mathbb{R}^4_{\geq 0} |, S + V + I + R = N\}.$$

Proof. We proof this theorem using a same approach with authors in [53,54]. System (1) can be rewritten as below:

$$\frac{dX}{dt} = CX + D, \tag{2}$$

where

$$X = (S, V, I, R)^T, \quad C = \begin{bmatrix} -d_1 & \theta & 0 & \delta \\ u & -d_2 & 0 & 0 \\ \beta I & 0 & -d_3 & \epsilon\beta I \\ 0 & 0 & \gamma & -d_4 \end{bmatrix},$$

with $d_1 = (\beta I + \mu + u)$, $d_2 = (\theta + \mu)$, $d_3 = (\gamma + \mu)$, $d_4 = (\epsilon\beta I + \delta + \mu)$ and $D = (A, 0, 0, 0)^T$. Because vector D is positive and all entries are not diagonal of C is non-negative. Thus, the matrix C is classified as a Metzler matrix for all $X \in \mathbb{R}^4_{\geq 0}$. So system (2) is positive invariant in $\mathbb{R}^4_{\geq 0}$. Therefore, system (1) from its initial state to completion remains in $\mathbb{R}^4_{\geq 0}$. □

3.2. Non-dimensionalization of variables and parameters

Before analyzing the model, a non-dimensionalization process of the population is carried out. Assuming that the number of natural births and death rates are the same, we have $\frac{dN}{dt} = 0$ consequently $A = \mu N$, such that $N = \frac{A}{\mu}$. this shows that the total population is constant.

Let $x_1 = \frac{S}{N}$; $x_2 = \frac{V}{N}$; $x_3 = \frac{I}{N}$, $x_4 = \frac{R}{N}$, $t\gamma = \tau$; $\frac{\mu}{\gamma} = m$; $\frac{\beta N}{\gamma} = b$; $\frac{\delta}{\gamma} = c$; $\frac{u}{\gamma} = u$; $\frac{\epsilon}{\gamma} = e$, $\frac{\theta}{\gamma} = d$ and $x_2 = 1 - x_1 - x_3 - x_4$, then system (1) can be reduced in to the following three dimensional system of differential equation.

$$\frac{dx_1}{d\tau} = m - bx_1x_3 + cx_4 - mx_1 - ux_1 + d(1 - x_1 - x_3 - x_4), \tag{3a}$$
$$\frac{dx_3}{d\tau} = bx_1x_3 + bex_3x_4 - x_3 - mx_3, \tag{3b}$$
$$\frac{dx_4}{d\tau} = x_3 - bex_3x_4 - cx_4 - mx_4. \tag{3c}$$

With this non-dimensionalization process, we have system in (1) reduced into a three dimensional system with all parameters and variables are dimensionless.

3.3. Equilibrium points and the reproduction number

Equilibrium point analysis is a crucial tool in epidemiological modeling as it provides insights into the long-term behavior and stability of the system. Equilibrium points, also known as steady states or fixed points, represent conditions where the dynamic of human (represented by the model variables) remain constant over time. In epidemiological models, equilibrium points typically correspond to disease-free states or endemic equilibrium where the disease persists at a constant level. There are some benefit on calculating equilibrium points, such as it will determine the disease endpoints (disease persist or extinct), stability analysis will helps researcher to determine whether small perturbations will change the possible outcome of interventions, and it also can help public health authorities to prepare some interventions to avoid possible endemic condition. Based on this, it is necessary to analyze the existence and stability of the equilibrium points of system (3), and how they may related to the reproduction number of the model.

The equilibrium points of system (3) is calculated by set the right hand side of system (3) equal to zero, and solve it respect to x_1, x_3, and x_4. The dimensionless tuberculosis model in system (3) has two types of equilibrium points, i.e:

1. **The TB-free equilibrium point.** This equilibrium represent a condition where no infected individuals in the population and given by

$$E_1 = (x_1^\dagger, x_3^\dagger, x_4^\dagger) = \left(\frac{m+d}{m+u+d}, 0, 0\right).$$

2. **The TB-endemic equilibrium point.** This equilibrium represent a condition when infected individuals exist in the population and given by

$$E_2 = \left(x_1^*, x_3^*, x_4^*\right) = \left(\frac{bemx_3^* + cm + m^2 + c + m}{(bex_3^* + c + m)b}, x_3^*, \frac{x_3^*}{bex_3^* + c + m}\right), \tag{4}$$

where x_3^* is taken from the positive roots of the following polynomial.

$$f(x_3) := a_2 x_3^2 + a_1 x_3 + a_0 = 0,$$

with $a_2 = -b^2 e(d+m)$, $a_1 = b(be(m+d) - [em(d+m+u) + +cd + d + m + m(m+d+c)]$, and $a_0 = (c+m)\left[b(m+d) - (1+m)(d+m+u)\right]$.

To analyze the existence and local stability of equilibrium points, we derive the reproduction number of system (3). The basic reproduction number represent the expected number of secondary cases due to one primary cases in a perfectly susceptible population during one infection period. The basic reproduction number of system (3) derived using the Next-Generation matrix approach [55,56]. In many epidemiological models, this reproduction number plays an important role as endemic indicator [57–59]. Most of these models state that the disease will persist if $\mathscr{R}_c > 1$ and extinct if $\mathscr{R}_c < 1$. In the context of tuberculosis, $\mathscr{R}_c$ serves as a critical epidemiological parameter that quantifies the transmission potential of the disease. It provides valuable insights into the dynamics of tuberculosis spread and informs public health efforts aimed at controlling and preventing the disease. The Basic reproduction number of system (3) is given by

$$\mathscr{R}_c = \frac{b(d+m)}{(d+m+u)(1+m)}. \tag{5}$$

In the concept of tuberculosis models considering the impact of reinfection on the effectiveness of vaccination strategies in controlling the spread of infection, $\mathscr{R}_c$ represents the expected number of secondary infections caused by a single infection during the infectious period in a susceptible population. The reproduction number plays a crucial role in determining the endemic nature of a disease in a population in numerous mathematical models. In our model, the significance of the reproduction number in determining the stability of the TB-free equilibrium is expressed through the following theorem.

Theorem 2. *The TB-free equilibrium point E_1 is locally asymptotically stable if $\mathscr{R}_c < 1$ and unstable if $\mathscr{R}_c > 1$*

Proof. The local stability criteria of TB-free equilibrium is obtained by analyzing the eigenvalues of the Jacobian matrix of system (3). The characteristic equation of this matrix is given by

$$(\lambda + d + m + u)(\lambda + c + m)\left(\lambda + 1 - \frac{b(d+m)}{d+m+u} + m\right) = 0, \tag{6}$$

where $\lambda_1 = -(d+m+u)$, $\lambda_2 = -(c+m)$, and $\lambda_3 = -(m+1)(1-\mathscr{R}_c)$. It is clear that λ_1 and λ_2 is always negative, while λ_3 is negative if and only if $\mathscr{R}_c < 1$. Hence the proof is completed. □

Next, we analyze the existence and local stability of the TB-endemic equilibrium point E_2. Expression of polynomial for E_2 in (4) can be rewritten as a function of $\mathscr{R}_c$ as follows by substituting $b = \dfrac{\mathscr{R}_c(d+m+u)(m+1)}{(d+m)}$:

$$f(x_3) := c_2 x_3^2 + c_1 x_3 + c_0 = 0,$$

where

$$\begin{aligned}
c_2 &= \frac{(1+m)^2(d+m+u)^2 e}{(d+m)}\mathscr{R}_c^2, \\
c_1 &= \frac{(1+m)(m+d+u)}{(d+m)}\mathscr{R}_c\left[(m+c+1)(m+d)\right. \\
&\quad \left. -\ e(m+d+u)((\mathscr{R}_c - 1)m + \mathscr{R}_c)\right], \\
c_0 &= (m+1)(m+c)(m+d+u)(1-\mathscr{R}_c).
\end{aligned}$$

It is clear to see that since $c_0 < 0 \iff \mathscr{R}_c > 1$, then $f(x_3)$ will have unique positive roots for $\mathscr{R}_c > 1$. Using implicit differentiation, then we have $\dfrac{\partial x_3}{\partial \mathscr{R}_c}(x_3 = 0, \mathscr{R}_c = 1) = -\dfrac{c_0}{c_1}(\mathscr{R}_c = 1) = -\dfrac{(m+c)(m+d)}{(m+c+1)(m+d) - e(m+d+u)}$. Therefore, we will have at least one positive root for $\mathscr{R}_c < 1$ if $\dfrac{\partial x_3}{\partial \mathscr{R}_c} < 0 \iff \mathcal{K} = (m+c+1)(m+d) - e(m+d+u) < 0$. Furthermore, since $f(x_3)$ is a second degree polynomial which is always continue in $x_3 \in \mathbb{R}$, then we will have a fold point of $f(x_3)$ at $\mathscr{R}_c$ which satisfy $c_1^2 - 4c_2c_0 = 0$. We call this $\mathscr{R}_c$ as $\mathscr{R}_c^*$. Based on this analysis, we have the following theorem.

Theorem 3. *Let $\mathscr{R}_c = \frac{b(d+m)}{(d+m+u)(1+m)}$, $\mathcal{K} = (m+c+1)(m+d) - e(m+d+u)$ and $\mathscr{R}_c^*$ is $\mathscr{R}_c$ that satisfy $c_1^2 - 4c_2c_0 = 0$. If $\mathcal{K} > 0$, then system (3) has:*

1. *No endemic equilibrium at $\mathscr{R}_c < 1$.*
2. *Unique endemic equilibrium at $\mathscr{R}_c > 1$.*

On the other hand, if $\mathcal{K} < 0$, then system (3) has:

1. *No endemic equilibrium at $\mathscr{R}_c < \mathscr{R}_c^*$.*
2. *Unique endemic equilibrium at $\mathscr{R}_c \geq 1$.*
3. *Two equal endemic equilibrium at $\mathscr{R}_c = \mathscr{R}_c^*$.*
4. *Two different endemic equilibrium at $\mathscr{R}_c \in (\mathscr{R}_c^*, 1)$.*

3.4. Bifurcation analysis

Next, we will analyze the local stability of the TB-endemic equilibrium around $\mathscr{R}_c = 1$ using the Castillo–Song bifurcation theorem [60].

Theorem 4. *The tuberculosis model (3) undergoes a backward bifurcation phenomenon at $\mathscr{R}_c = 1$ if $\mathcal{K} < 0$ and forward bifurcation when $\mathcal{K} > 0$.*

Proof. Use branch from system (1) The one that has already been dimensionless becomes the system (3). Let $\frac{dx_1}{d\tau} = f_1$, $\frac{dx_3}{d\tau} = f_2$, $\frac{dx_4}{d\tau} = f_3, x_1 = y_1, x_3 = y_2, x_4 = y_3$, so that system(3) can be rewritten as:

$$\begin{aligned}
f_1 &= m - by_1y_2 + cy_3 - my_1 - uy_1 + d(1 - y_1 - y_2 - y_3), \\
f_2 &= by_1y_2 + bey_2y_3 - y_2 - my_2, \\
f_3 &= y_2 - bey_2y_3 - cy_3 - my_3.
\end{aligned}$$

By setting $\mathscr{R}_c = 1$, we obtain the bifurcation parameter $b = b^*$ where $b^* = \dfrac{(d+m+u)(1+m)}{(d+m)}$.

Furthermore, the linearization matrix around the TB-free equilibrium point (DFE) for $b = b^*$ is obtained as follows:

$$J = \begin{bmatrix} -d-m-u & -1-m-d & c-d \\ 0 & 0 & 0 \\ 0 & 1 & -c-m \end{bmatrix}. \tag{7}$$

The eigen value of Jacobian of the system (7), evaluated at (4) with $b = b^*$, are given by: $\lambda_1 = -(d+m+u)$, $\lambda_2 = 0$, and $\lambda_3 = -(c+m)$. Thus

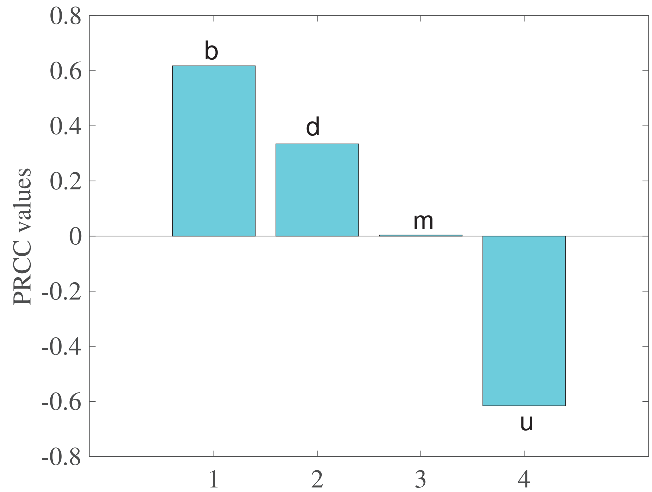

Fig. 2. Tornado plots showing the sensitivity analysis of model parameters on $\mathscr{R}_c$.

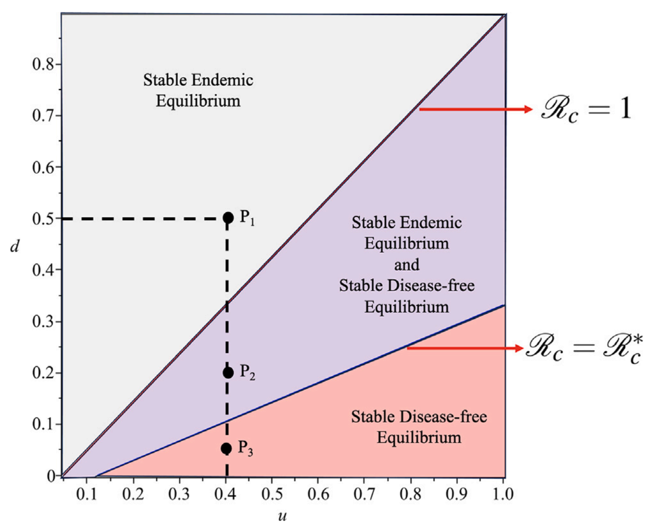

Fig. 3. Area of stable equilibrium of system (3) depend on the combination of u and d. Parameter values are $b = 2.16,\ c = 3/7,\ m = 3/65, e = 2$.

$\lambda_2 = 0$ is a simple zero eigen value and the other eigen values are the real and negative.

Next, equation (7) has a right eigen vector associated with the zero eigen value $\lambda_2 = 0$ are given by $w = (w_1, w_2, w_3)^T$, where $w_1 = -\frac{(d+m)(c+m+1)w_3}{(d+m+u)}; w_2 = (c+m)w_3$, and $w_3 = w_3 > 0$, so that $w = (-\frac{(d+m)(c+m+1)}{(d+m+u)}, (c+m), 1)^T$. Further, a left eigen vector associated with the zero eigen value $\lambda_2 = 0$ are given by $v = (v_1, v_2, v_3)$, where $v_1 = 0; v_2 = v2 > 0; v_3 = 0$, so that $v = (0, 1, 0)$.

The associated non zero partial derivatives of (f_k) are given by:

$$\frac{\partial^2 f_1}{\partial y_1 \partial y_2} = \frac{\partial^2 f_1}{\partial y_2 \partial y_1} = -b, \qquad \frac{\partial^2 f_2}{\partial y_1 \partial y_2} = \frac{\partial^2 f_2}{\partial y_2 \partial y_1} = b,$$
$$\frac{\partial^2 f_2}{\partial y_2 \partial y_3} = \frac{\partial^2 f_2}{\partial y_3 \partial y_2} = be, \qquad \frac{\partial^2 f_3}{\partial y_2 \partial y_3} = \frac{\partial^2 f_3}{\partial y_3 \partial y_2} = -be.$$

We chose b as the bifurcation parameter and it can be shown that the associated partial derivatives of (f_k) are:

$$\frac{\partial^2 f_1}{\partial y_2 \partial b} = -\frac{d+m}{d+m+u}, \qquad \frac{\partial^2 f_3}{\partial y_3 \partial b} = \frac{d+m}{d+m+u}.$$

As a result of the second partial derivative, the coefficients $\mathcal{A}$ and $\mathcal{B}$ defined and computed as in the following:

$$\mathcal{A} = \sum_{k,i,j=1}^{3} v_k w_i w_j \frac{\partial^2 f_k}{\partial y_i \partial y_j}(0,0) = \frac{-(d+m+u)\,e + (c+m+1)\,(d+m)}{d+m+u}. \tag{8}$$

$$\mathcal{B} = \sum_{k,i,j=1}^{3} v_k w_j \frac{\partial^2 f_k}{\partial y_i \partial b}(0,0) = \frac{(c+m)\,(d+m)}{d+m+u}. \tag{9}$$

From equations (8) and (9), we can show, $\mathcal{B}$ always has a positive sign. On the other hand, $\mathcal{A}$ is positive when $\mathcal{K} > 0$ and negative when $\mathcal{K} < 0$ where $\mathcal{K} = (m+c+1)(m+d) - e(m+d+u)$. Therefore, Model (3) undergoes backward bifurcation when $\mathcal{K} > 0$, and forward bifurcation when $\mathcal{K} < 0$. Hence the proof is complete. □

4. Numerical experiments

In this section, we present the results of numerical experiments conducted on the model. The first part involves a mathematical analysis to gain a better understanding of how the basic reproduction number varies with changes in the parameters.

4.1. Sensitivity and elasticity analysis

At this stage an analysis will be carried out to examine the sensitivity of the control reproduction number $\mathscr{R}_c$ to various parameters that affect its value, as well as the relationship between these parameters. The purpose of conducting a sensitivity analysis in the context of our tuberculosis model is to assess the robustness and reliability of our model outcomes in response to variations or uncertainties in key input parameters. By systematically varying the values of these parameters within plausible ranges, sensitivity analysis allows us to identify which parameters have the most significant impact on the model outcomes and to quantify the extent of their influence. By considering the different values for these parameters, we can observe how the control reproduction number of the system (3) varies while keeping the values of the other parameters constant. We conduct two types of sensitivity analysis on the reproduction number, namely with elasticity index approach [61] and global sensitivity analysis with Latin Hypercube Sampling (LHS)/Partial Rang Correlation Coefficient method [62].

To evaluate the sensitivity, we employ the forward sensitivity index, which is normalized with respect to the reproduction control $\mathscr{R}_c$ and is denoted as

$$C^p_{\mathscr{R}_c} = \frac{\partial \mathscr{R}_c}{\partial p} \times \frac{p}{\mathscr{R}_c},$$

where p is any parameters in $\mathscr{R}_c$. Using above formula, we have the following results:

$$C^b_{\mathscr{R}_c} = 1, \quad C^d_{\mathscr{R}_c} = \frac{\partial \mathscr{R}_c}{\partial d} \times \frac{d}{\mathscr{R}_c} = \frac{bu}{(d+m+u)^2(m+1)},$$
$$C^c_{\mathscr{R}_c} = \frac{\partial \mathscr{R}_c}{\partial c} \times \frac{c}{\mathscr{R}_c} = 0, \quad C^e_{\mathscr{R}_c} = \frac{\partial \mathscr{R}_c}{\partial e} \times \frac{e}{\mathscr{R}_c} = 0,$$
$$C^m_{\mathscr{R}_c} = \frac{\partial \mathscr{R}_c}{\partial m} \times \frac{m}{\mathscr{R}_c} = -\frac{b(d^2 + (2m+u)d + m^2 - u)}{(d+m+u)^2(m+1)^2},$$
$$C^u_{\mathscr{R}_c} = \frac{\partial \mathscr{R}_c}{\partial u} \times \frac{u}{\mathscr{R}_c} = -\frac{b(d+m)}{(d+m+u)^2(m+1)}.$$

To calculate the sensitivity index, we using parameter values as in Fig. 5 and 10 different values of u. The value of elasticity index for each parameter is given by the average of this 10 examples and given in Table 2.

From Table 2 it can be seen that all parameters have influence. Sensitivity analysis revealed that the parameter with the highest sensitivity was vaccination to susceptible individuals, represented by u. A positive sensitivity index indicates that $\mathscr{R}_c$ increases with an increase

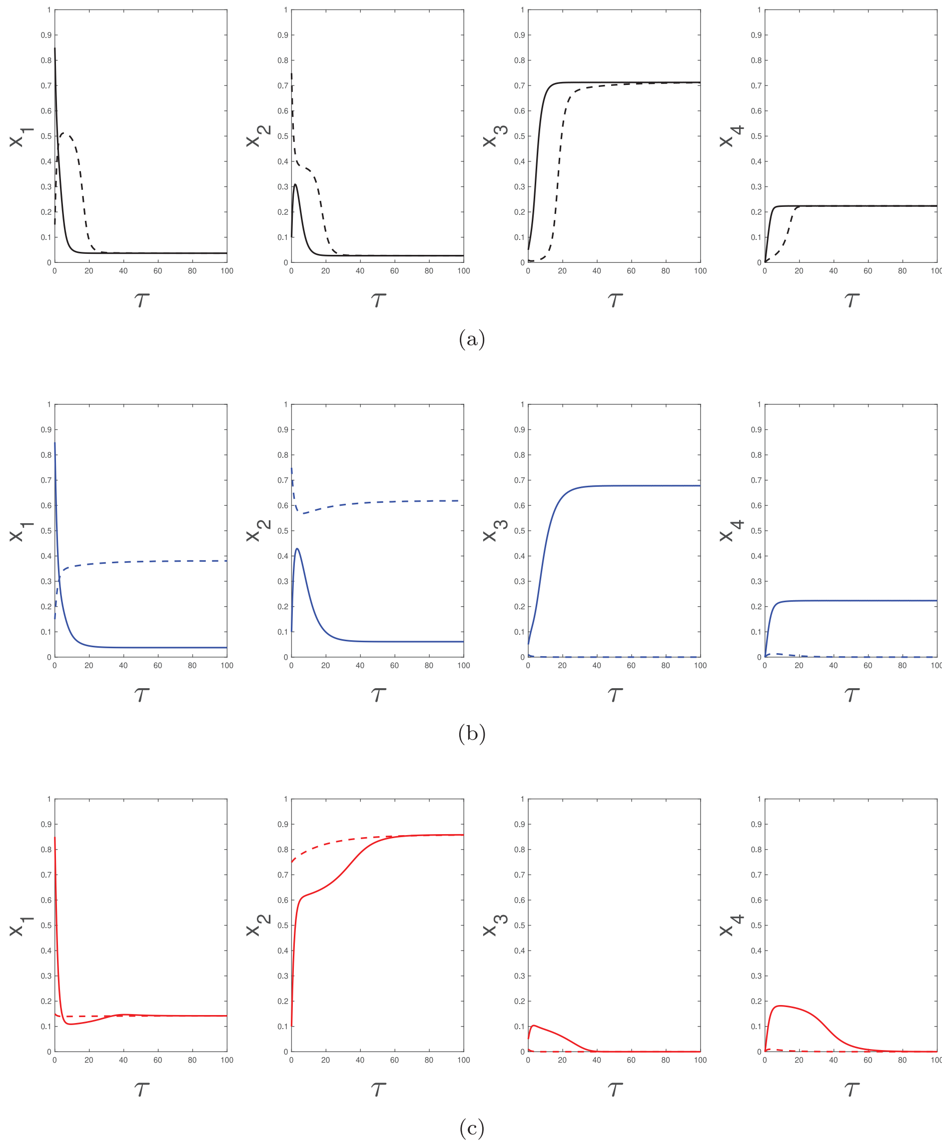

Fig. 4. Solution of system (3) depend on the combination of u and d based on Fig. 3. Parameter values are $b = 2.16$, $c = 3/7$, $m = 3/65, e = 2, u = 0.4$, while d varying, namely 0.5 for panel (a), 0.2 for panel (b), and 0.03 for panel (c).

Table 2
Sensitivity index $\mathscr{R}_c$ with respect to the model parameters. Parameter values are $b = 2.16$, $c = 3/7$, $m = 3/65, d = 1/5, e = 2$, while u was taken from 10 different parameter values.

Parameter	m	b	d	u	c	e
Sensitivity index	+0.054	+1	+0.424	−0.522	0	0

in this parameter, while a negative sensitivity index indicates that $\mathscr{R}_c$ decreases with an increase in u. In summary, $\mathscr{R}_c$ is increasing due to factors such as increased transmission of tuberculosis to humans (b), natural death rates (m), and reduced effectiveness of control measures via vaccination (d). Conversely, $\mathscr{R}_c$ decreases with an increase in the number of susceptible individuals vaccinated (u). The sensitivity of $\mathscr{R}_c$ with respect to u is -0.522, indicating that a 1% increase in u will cause a 0.522% decrease in $\mathscr{R}_c$. Based on these findings, we recommend adopting effective tuberculosis control measures, particularly increasing individual vaccination rates (u), as the most impactfull strategy for reducing $\mathscr{R}_c$.

The next step in our analysis involves conducting a global sensitivity analysis on the basic reproduction number ($\mathscr{R}_c$), the results of which are presented in Fig. 2. The model parameters exhibit inherent uncertainties due to variations influenced by factors such as geographical location, demographic characteristics, and social behavior. This variability is reflected in the range of possible parameter values presented in Table 1. To address these uncertainties, we employ the Latin Hypercube Sampling (LHS) technique for uncertainty quantification and sensitivity analysis. To assess the relationship between input and output variables, we utilize the Partial Rank Correlation Coefficient (PRCC) [62]. Parameters with PRCC values ranging between 0.5 and 1, or between 0 and −1, are deemed most significant. Our analysis reveals that parameters b and u, representing infection and vaccination scaling factors, respectively, exert the most substantial influence on $\mathscr{R}_c$. Specifically, an increase in b leads to an increase in $\mathscr{R}_c$, while an increase in u results in a reduction of $\mathscr{R}_c$. Based on these findings, strategies aimed at reducing the infection rate and increasing the vaccination rate are crucial for mitigating the spread of tuberculosis, as indicated by lower values of $\mathscr{R}_c$.

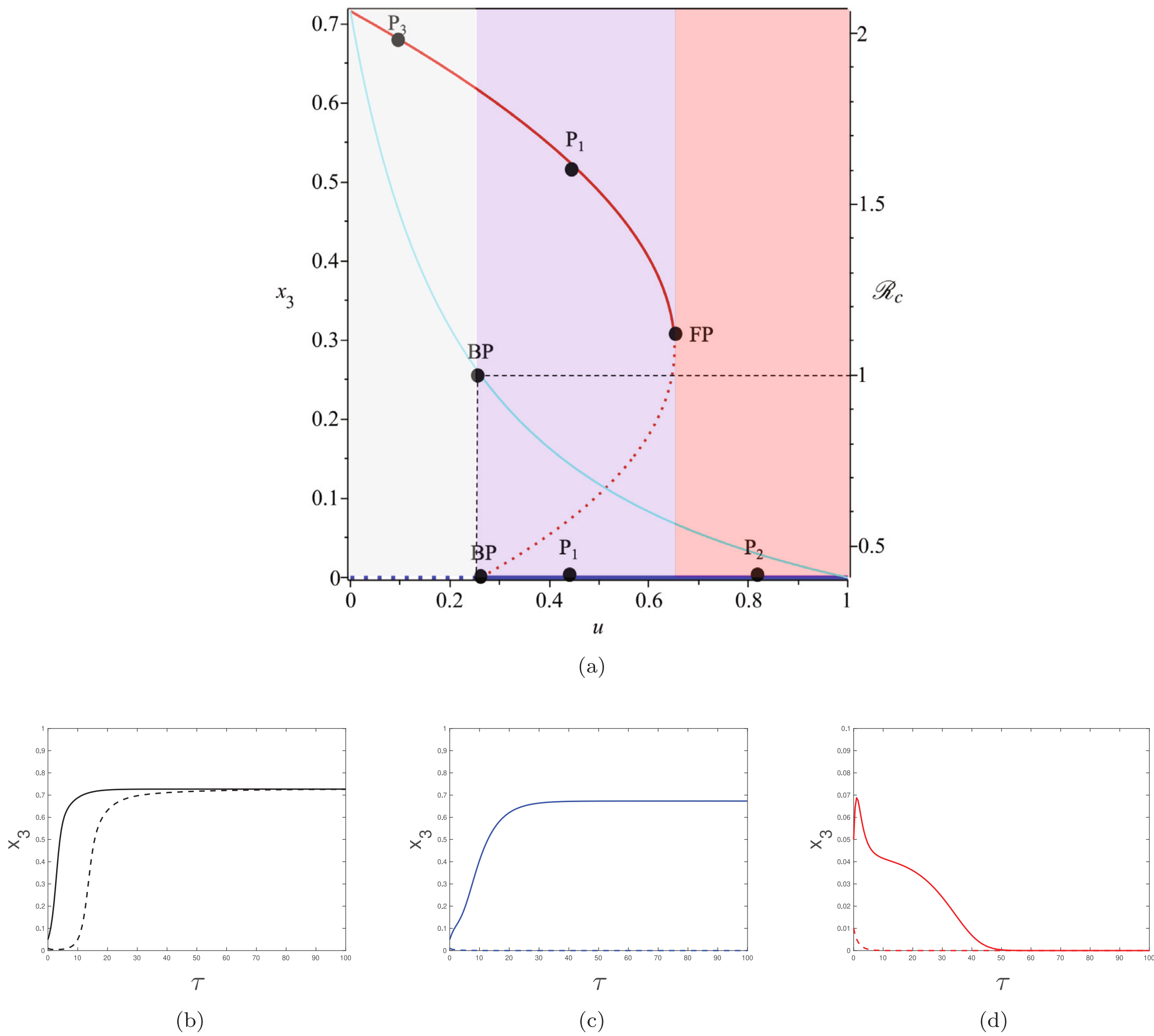

Fig. 5. (a) Backward bifurcation diagram of system (3) respect to u. The red and blue curves represent endemic and disease free equilibrium, respectively. Dash and dot curves represent stable and unstable equilibrium, respectively. Cyan curve represents $\mathscr{R}_c$ as a function of u. Parameter values are $b = 2.16$, $c = 3/7$, $m = 3/65$, $d = 1/5$, $e = 2$. Panel (b), (c), and (d) show a solution of x_3 respect to system (3) for parameter values $u = 0.085$ at P_3, $u = 0.43$ at P_1 and $u = 0.81$ at P_2, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4.2. Bifurcation analysis

In this section, we analyze the impact on the combination between vaccination (u) and it effective duration (d). As we already mentioned in the previous section, vaccination u has a negative elasticity index, which means that increasing u will reduce the reproduction number $\mathscr{R}_c$. On the other hand, vaccine duration d has a positive index of elasticity. This means that shorter duration of vaccine (smaller d) will reduce $\mathscr{R}_c$. From example that we give in Table 2, we see that the rate of vaccine implementation in the population is more sensitive compared to it valid duration.

Now, we analyze the impact of these two parameters on determining the disappearance of tuberculosis from the population. To understand this, we visualize Theorem 4. The results are given in Fig. 3 with the parameter values given in it caption. With this combination of parameter values, backward bifurcation will appear. Hence, We have $\mathcal{K} < 0$. The gray area is when the combination between u and d does not succeed to eradicate tuberculosis from the population. In this area, we have $\mathscr{R}_c > 1$. The red area represents a possible combination between u and d that will always succeed to eradicate tuberculosis. We have $\mathscr{R}_c < \mathscr{R}_c^*$ in this area. On the other hand, the purple area represents a combination between u and d that will not always succeed to eliminate tuberculosis from the population., since we have $\mathscr{R}_c^* < \mathscr{R}_c < 1$. If the duration of the tuberculosis vaccine to be effective is not long enough, then the condition of $\mathscr{R}_0 < 1$ is not enough to guarantee the disappearance of tuberculosis. To illustrate the stable equilibrium point in each area, we conduct an autonomous simulation of system (3) by choosing three sample points in each area in Fig. 3, namely $P_1(0.4, 0.5)$, $P_2(0.4, 0.2)$, and $P_3(0.4, 0.03)$. The results are given in Fig. 4. We can see that panel (a) and (c) show all initial condition tends to one stable equilibrium point, namely the endemic and TB-free equilibrium point, respectively. On the other hand, panel (b) show that the solution of system (3) depends on its initial condition. Panel (b) shows two stable equilibrium appears, namely the endemic and TB-free equilibrium points.

Next, we conduct simulations to describe two possible bifurcations of system (3) at $\mathscr{R}_c = 1$. Firstly, we illustrate the backward bifurcation phenomenon when $\mathcal{K} < 0$, as shown in Fig. 5, panel (a). In this simulation, we identify the Branching Point (BP) at $\mathscr{R}_c = 1$, reached at $u = 0.262$. As the vaccination rate u increases, a stable TB-free equilibrium point emerges, presenting an opportunity for tuberculosis elimination within the population. However, reducing the vaccination rate diminishes this chance. Between the branching point and the fold point, located at $u \in (0.262, 0.651)$ in Fig. 5 panel (a), two stable equilibrium

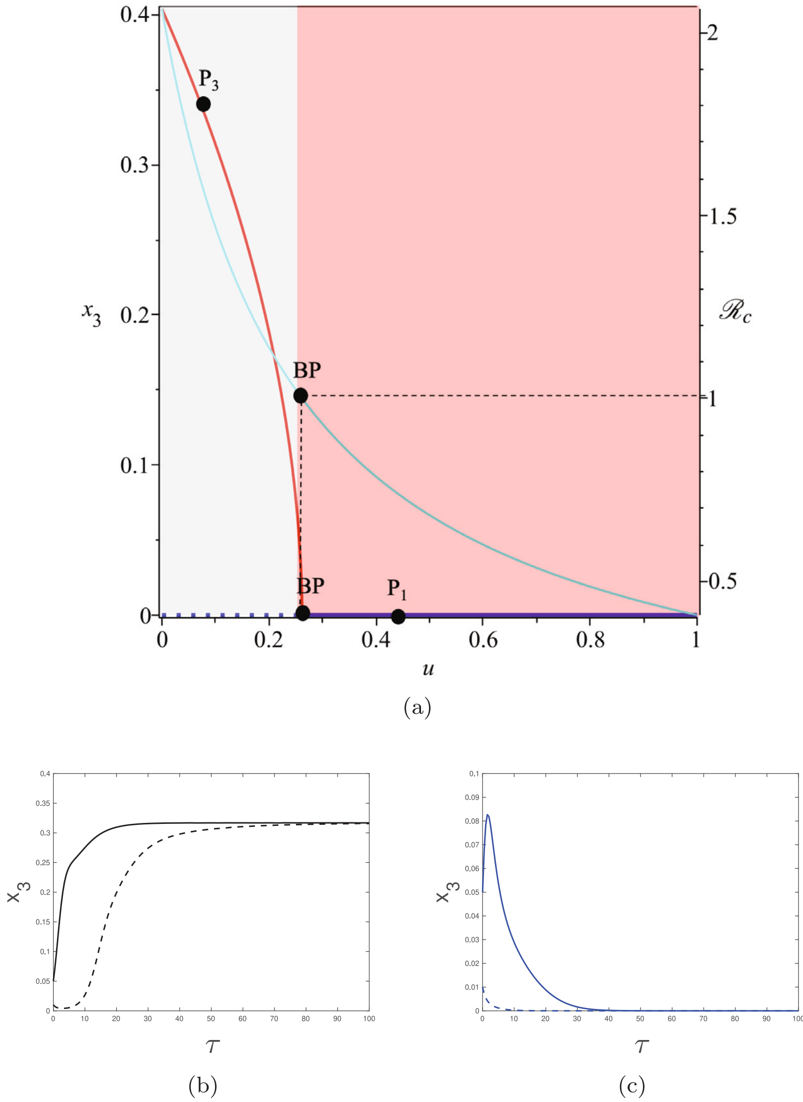

Fig. 6. Forward bifurcation diagram of system (3) respect to u. Red and blue curve represent endemic and disease-free equilibrium, respectively. Dash and dot curves represent stable and unstable equilibrium, respectively. Cyan curve represents $\mathscr{R}_c$ as a function of u. Parameter values are $b = 2.16$, $c = 3/7$, $m = 3/65$, $d = 1/5$, $e = 0.7$. Panel (b) and (c) show a solution of x_3 respect to system (3) for parameter values $u = 0.085$ at P_3, and $u = 0.43$ at P_1, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

points coexist: the TB-free and TB-endemic equilibrium points. Notably, a bistability phenomenon occurs within this region (purple area). As u decreases below the branching point, the TB-free equilibrium point becomes unstable, while the TB-endemic equilibrium point remains stable. Panels (b), (c), and (d) depict autonomous simulations of three sample points (P_1, P_2, P_3) from each area in Fig. 5 panel (a). Panels (b) and (d) demonstrate the convergence of all initial conditions towards the endemic and TB-free equilibrium points, respectively (sample point P_3 and P_2). Conversely, panel (c) showcases the system's trajectory towards different equilibrium points due to bistability phenomena.

In the next simulation, we change the reinfection parameter e from $e = 2$ in Fig. 5 to $e = 0.7$ in Fig. 6. With $e = 0.7$, then we have $\mathcal{K} > 0$, which will trigger a forward bifurcation phenomena at $\mathscr{R}_c = 1$. Hence, bistability phenomena does not appears. We run the solution of x_3 with a same parameter values P_1 and P_3 as in Fig. 5, and the results are given in panel (a) and (b). For a sample points P_3, which is the same as in Figs. 5 and 6, we can see that the peak endemic equilibrium size of P_1 when forward bifurcation appears is smaller than when backward bifurcation appears. Therefore, we can conclude that backward bifurcation may trigger a high endemic size at $\mathscr{R}_c > 1$ compared to when forward bifurcation appears. In the next simulation in Fig. 7, we show the impact of reinfection parameter e to the type of bifurcation phenomena at $\mathscr{R}_c = 1$ by setting all parameter values are the same, except e varying from 0 to 2.5. We can see that a larger value of e will increase the chance for the appearance of backward bifurcation. Furthermore, the fold point when backward bifurcation appears getting larger when e also larger. Fig. 8 show the impact of u on the solution of tuberculosis model in system (3). It is clear to see that larger rate of vaccination given to susceptible individuals will increase the proportion of susceptible and vaccinated individuals, and reducing the proportion of infected and recovered individuals.

5. Conclusion

In this paper, a tuberculosis transmission model has been developed, taking into account the impact of reinfection and the effectiveness of vaccination strategies. The model has two equilibrium points: the TB-free equilibrium and the endemic equilibrium. The analysis using the

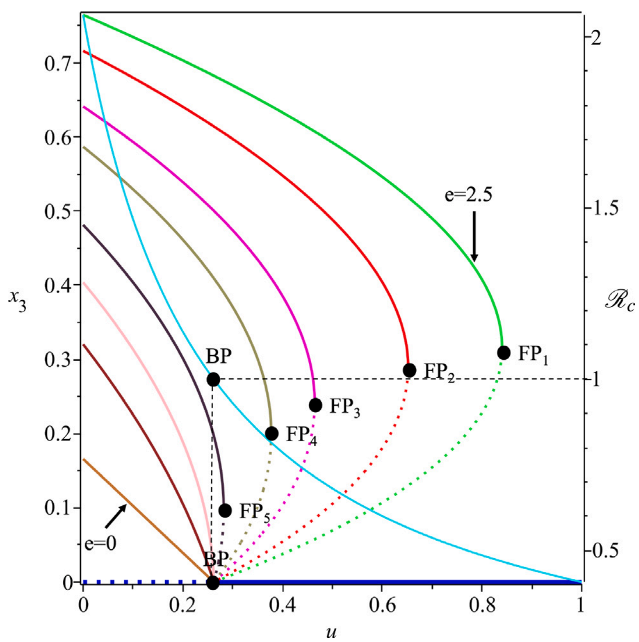

Fig. 7. Impact of e on the type of bifurcation of system (3) at $\mathscr{R}_c = 1$. FP and BP is the fold and branch point, respectively. Parameter values are $b = 2.16$, $c = 3/7$, $m = 3/65$, $d = 1/5$, while e varying from 0 to 2.5.

Next Generation Matrix (NGM) method yielded the control reproduction number ($\mathscr{R}_c$), and it was found that the TB-free equilibrium is locally asymptotically stable when $\mathscr{R}_c < 1$. Additionally, using the Castillo–Song bifurcation theorem, it was determined that the tuberculosis model exhibits backward bifurcation at $\mathscr{R}_c = 1$ if $\mathcal{K} < 0$ and forward bifurcation when $\mathcal{K} < 0$. With the appearance of this backward bifurcation, it means that it is still possible to have an endemic of tuberculosis even though $\mathscr{R}_c < 1$. This backward bifurcation was triggered by the reinfection coefficient. The larger value of reinfection will increase the chance of backward bifurcation to appears. The presence of a backward bifurcation induced by reinfection poses several challenges and limitations in the practical utility of the model for guiding public health strategies. One notable challenge as mentioned before is that reinfection does not explicitly appear in the formula for the control reproduction number, which serves as a key indicator of endemicity. This omission can lead to misguidance when $\mathscr{R}_c$ is solely relied upon as an indicator of disease prevalence. Furthermore, the occurrence of a backward bifurcation means that even when $\mathscr{R}_c < 1$, a large-scale endemic condition can persist. This challenges the traditional understanding that reducing $\mathscr{R}_c$ below unity is sufficient for disease elimination. Instead, the presence of reinfection-induced backward bifurcation highlights the need for additional interventions beyond reducing transmission rates to effectively control disease spread.

Furthermore, sensitivity and elasticity analyses were conducted on the model parameters. In our research, faced with the difficulty of accessing TB incidence data regularly, we resorted to an extensive review of literature to guide our parameter selection. These parameters were depicted as ranges to accommodate inherent fluctuations as we have shown in Table 1. To delve deeper into understanding the influence of crucial parameters on our model's results, we conducted sensitivity analysis utilizing the Latin Hypercube Sampling/Partial Rank Correlation Coefficient (LHS/PRCC) method. This enabled us to methodically adjust parameter values within their designated ranges and assess their effects on model outcomes, specifically focusing on the control reproduction number ($\mathscr{R}_c$). The parameters b, m, and d were found to be positively related to $\mathscr{R}_c$, meaning that an increase in these parameters leads to an increase in $\mathscr{R}_c$ and vice versa. On the other hand, the parameter u was negatively related to $\mathscr{R}_c$, indicating that an increase in u results in a decrease in $\mathscr{R}_c$ and vice versa. To effectively control tuberculosis transmission, it is essential to maintain a high level of vaccination, which is crucial in curbing the spread and reinfection of tuberculosis disease. A longer period of vaccination to protect people from tuberculosis, the smaller chance to have tuberculosis in the population. Based on these analysis, our findings underscore the potential effectiveness of vaccination strategies in reducing TB transmission under ideal conditions. By achieving high vaccination coverage, policymakers can significantly reduce the incidence of primary TB infections and contribute to overall disease control efforts. However, our study also highlights the importance of considering reinfection dynamics in TB control strategies. Despite the efficacy of vaccination in reducing primary infections, reinfection rates can sustain TB transmission and undermine the long-term success of control measures. Therefore, policymakers and public health practitioners should adopt comprehensive TB control strategies that address both primary infections and reinfection dynamics.

While our model incorporates key factors such as vaccination and reinfection dynamics, there are potential scenarios and contexts where our findings may not be directly applicable or where further refinement of the model is necessary to accurately represent TB dynamics. One notable limitation is the absence of consideration for individuals with multi-drug-resistant tuberculosis (MDR-TB) within the model, as many previous author already discuss [63,64]. This is an important consideration as MDR-TB poses unique challenges in treatment and transmission dynamics. Future refinements to the model could involve incorporating a separate compartment for MDR-TB individuals to better capture their impact on TB dynamics. Additionally, our model does not account for the possibility of TB coinfection with other diseases such as COVID-19 or HIV. Coinfections can significantly influence disease progression and treatment outcomes, requiring a more comprehensive modeling approach. Future iterations of the model could include mechanisms to simulate TB coinfection scenarios to better understand their implications for disease control strategies (please see [65,66] for some earlier work by other researchers or a more complex involving interaction between three disease [67]). Another possible direction for the research is by incorporating the genetically resistant human on the model. This is crucial considering they may contribute on the disease progression and affecting treatment outcomes. In summary, while our model provides valuable insights into TB dynamics, there are opportunities for further refinement to enhance its applicability in diverse contexts and scenarios. Addressing these limitations will contribute to a more comprehensive understanding of TB transmission dynamics and guide more effective public health interventions. Lastly, it is important incorporating empirical data through real-world calibration, demographic integration, and model validation which can enhance the realism and predictive power of the TB transmission model, providing valuable insights for guiding public health interventions and policy decisions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgments

We thank to the editor and the reviewers for their valuable comments and suggestions in enhancing the quality and clarity of this manuscript.

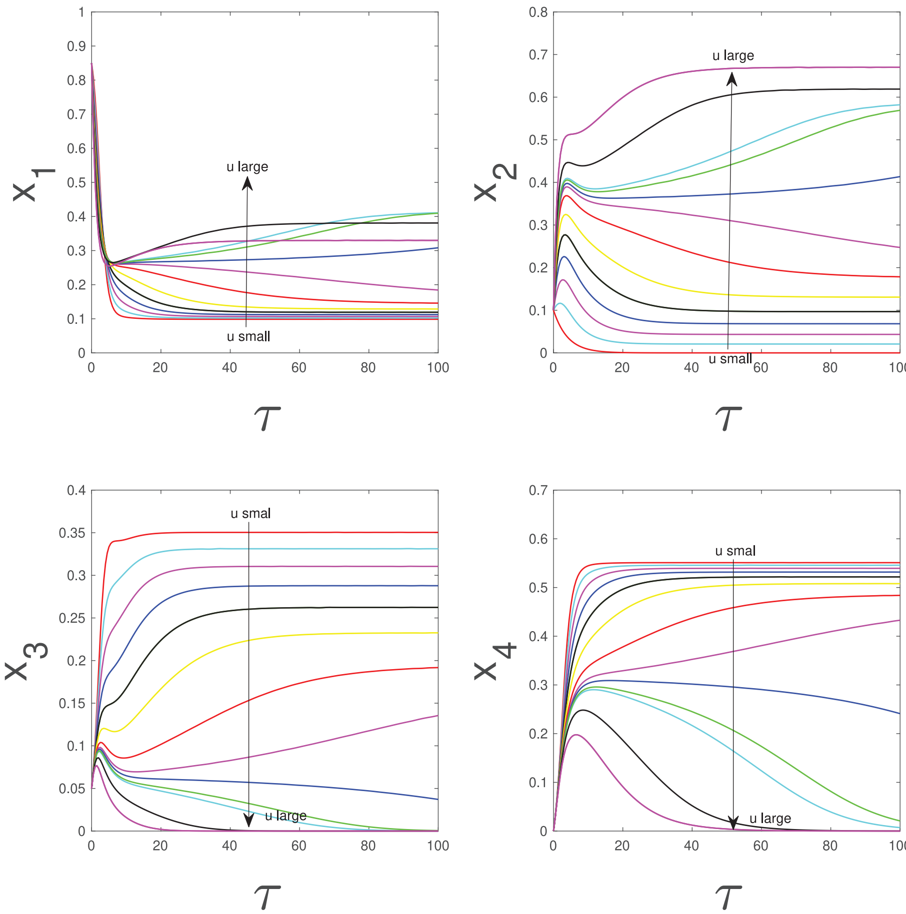

Fig. 8. Impact of u on the dynamic of x_i for $i = 1, 2, 3, 4$. Parameter values are $b = 2.16$, $c = 3/7$, $m = 3/65$, $d = 1/5, e = 0.7$, while u varying from 0 to 1.

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